

TACTICAL PIVOT POSITIONING SYSTEM

360° Range Calibration Panel

Quick User Guide

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v1.0

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01 Starting the App

This app finds the exact center point that your camera platform rotates around. You do this by pointing at the same fixed spot on the floor from several different angles — the app then works out the true center for you.

- 1 Open the application folder.
- 2 Double-click `CalibrationApp.exe` to launch it. No installation is required.
- 3 The main window opens with the video area on the left and the control panel on the right.

02 Connecting to the Camera

At the top of the right-hand panel, under **Operation Mode**, choose one of two options:

VIRTUAL SIMULATOR

Selected by default. Shows a practice floor — use this to try the app out with no camera connected.

LIVE RTSP FEED

Shows the real camera picture. Use this for an actual calibration.

To connect to the live camera

- 1 Select **Live RTSP Feed**.
- 2 In the **Stream Address String** box, type the camera's address, for example `rtsp://192.168.3.1/livestream`.
- 3 Click **Connect Stream**. The picture appears once the connection is made.
- 4 To stop, click the same button again (it now reads **Disconnect Stream**).

NOTE

If the button shows `Connect Stream (Error!)`, double-check that the camera is powered on, connected to the same network, and that the address was typed correctly.

03 Taking Calibration Points

Once you can see the picture (simulator or live), follow these steps to calibrate:

- 1 **Choose a fixed spot** on the floor that you can clearly see and identify — a tile corner, mark, or sticker works well.
- 2 **Rotate the camera** using the **Camera Rotation** slider in the panel (or by clicking and dragging on the picture itself).
- 3 **Move the crosshair onto that spot** — click and drag the yellow/cyan box in the middle of the picture directly on top of your chosen spot.
- 4 **Click “Take Reading”**. A cyan point (P1, P2, ...) is placed and logged in the **Calibration Log** list.
- 5 **Rotate to a new angle** and repeat steps 2–4, lining the crosshair up on the *same spot* each time. Take at least **4 points** in total, spread around different angles.
- 6 Once 4 or more points are logged, click **Calculate Center**. The app works out the true center and marks it on the picture with a green target labelled `CALIB_ZERO`.

TIP

Take your points spread out across the full turn (roughly every 60–90°) rather than all close together — this gives a much more accurate result.

04 Quick Tips

RESET CROSSHAIR TO CENTER

Puts the crosshair back in the middle of the screen if you lose track of it.

CLEAR ALL READINGS

Deletes all points and any calculated center so you can start over.

IF THIS HAPPENS...	DO THIS
The picture keeps saying "WAITING FOR RTSP VIDEO FEED..."	Check the camera is powered on and connected to the network, then try Connect Stream again.
"Calculate Center" button will not click	You need at least 4 points logged first — keep taking readings.
The calculated center looks off	Make sure every point was taken on the exact same physical spot, and spread across different angles rather than bunched together. Use Clear All Readings and try again.